

PARUL UNIVERSITY
FACULTY OF ENGINEERING & TECHNOLOGY
B.Tech. Summer 2023 - 24 Examination

Semester: 4
Subject Code: 303105210
Subject Name: Computer Organization and Microprocessor

Date: 27/05/2024
Time: 10:30am to 1:00pm
Total Marks: 60

Instructions:

1. All questions are compulsory.
2. Figures to the right indicate full marks.
3. Make suitable assumptions wherever necessary.
4. Start new question on new page.

Q.1	Objective Type Questions - (Fill in the blanks, one word answer, MCQ- not more than Five in case of MCQ) (All are compulsory) (Each of one mark)	(15)	CO	PO	Bloom's Taxonomy
	1. _____ is the highest priority Interrupt.		1	2	Analysis
	2. Which of the following is 16-bit instruction? A. LXI H, 0010H B. MOV D, A C. MVI C, A2H D. JNC X1		3	2	Understand
	3. Give example of branching instruction for 8085.		2	3	Understand
	4. Define: Opcode with example.		4	1	Knowledge
	5. Define Operand with example.		2	1	Ability
	6. What is Subroutine?		3	2	knowledge
	7. How many T-state is required for OUT 0A H Instruction?		3	4	understand
	8. _____ is an example of conditional loop instruction. A. JC B. INR C. DCR D. HLT		6	3	analysis
	9. _____ is used to read the content of IMR (Interrupt mask register). A. RIM B. SIM C. EI D. DI		3	4	Ability
	10. What is call instruction in 8085?		1	3	Ability
	11. _____ carries the data or address bits. A. Bus B. ALU C. Decoder D. ALE		6	2	analysis
	12. The operation executed on the data stored in the register are called as _____.		2	5	Ability and classification
	13. When ALE goes High, _____ bits is selected from AD0-AD7.		1	2	classification
	14. How many 16 bit registers are there in 8085?		5	4	Knowledge & classification
	15. Flag register is of _____ bit.		2	3	Understand

Q.2	Answer the following questions. (Attempt any three)	(15)			
	A) Which instruction is used to control interrupt. Give its types and explain any one with diagram.		4	2	Ability
	B) What is the use of register in 8085? How register is different from memory?		3	1	analysis
	C) Elaborate cache memory.		5	3	Knowledge & classification
	D) Describe following instruction in detail with example: I. LXI instruction II. LHLD III. STAX IV. CMA V. RRC		3	1	Ability
Q.3	A) Draw and explain de-multiplexed address data bus.	(07)	2	4	knowledge
	B) Write a program to divide two 8-bit data given as per memory location, Input: 7015H – 50H 7016H – 05H Store output at 5016H memory location.	(08)	6	3	understand
	OR				
	B) Define Machine cycle. List its types and draw Timing diagram for IO read Instruction.	(08)	5	2	analysis
Q.4	A) Describe what is an Interrupt? Explain each priority wise Interrupt types. Also describe EI and DI instruction?	(07)	2	3	Understand
	OR				
	A) Interface two memories RAM and EPROM having size of 8kbyte each. Find its starting address and Ending address.	(07)	3	1	Analysis
	B) Draw and explain any three blocks of architecture of 8085.	(08)	2	5	Knowledge & classification